WORK ORDER FOR GOOGLE AI STUDIO

IWACU RWANDA: ONE SITE, ONE PROJECT

Date: 2 November 2026 | Repository: eddyizere24-collab/Iwacu-rwanda-2 | Written by Claude (verifier) for the owner

Purpose. The owner wants a single real site. Your job is to make the one set of code changes listed here, all at once, in one working session. After you finish, other agents (Grok for CI and browser checks, Claude for verification) will run the tests and fix small mistakes. You do not need to run heavy tests and you must not touch the database.

How to read this order. Statements marked **VERIFIED** were read by Claude from the repository on 2 November 2026 (file name and line number given). Statements marked **CHECK** are things you must confirm yourself before acting. If a VERIFIED fact does not match what you see, STOP that task and report the difference. Never guess, never invent.

Order of work. T1 (own map), T2 (calibration), T3 (service worker), T4 (notes). Stop after T4. Everything else is listed in section 6 and is NOT yours.

1. Hard rules (read first)

1. **Branch.** Create the branch `studio/unify` from the branch `fix/consolidation-build`. Do NOT start from `main`: main currently fails to build on Vercel (error lint_or_type_error), the branch `fix/consolidation-build` builds. Never commit to main. If your GitHub export can only write to main, STOP, export nothing, and write: BLOCKED: can only write to main.
2. **Forbidden areas.** Do not touch the Supabase database in any way (no SQL, no migration, no table or function). No environment variables, no keys, no Vercel settings. Nothing under `.github/workflows/`, `legacy/`, `national-grid/`, `grid/`, `public/rwanda-grid-map/`. Do not edit `src/app/api/*` or `src/lib/locate-protection.ts`, `classifier-flow.ts`, `location-ranking.ts`, `location-prior.ts`, `location-hierarchy.ts`, `representative-location.ts`.
3. **No invented content.** No invented place names, villages, leaders, photos, phone numbers, percentages or test results. Every percentage shown to a citizen must come from the API response, never from your code.
4. **No new third-party hosts at runtime.** After your change the map may contact only its own domain and the existing own API routes. No CDN, no tile server, no font server, no image host.
5. **No new npm packages.** Only `bun.lock` exists in the repository root (VERIFIED) and Vercel builds with `bun run build` (VERIFIED). A new dependency would change the lock file and can break the build. Use the vendored files named in T1.
6. **Language.** All visible text stays in Kinyarwanda, in the same tone as the present text.
7. **Few commits.** Vercel builds every push and the free plan allows 100 deployments a day. Make at most 3 commits on the branch (one per task T1, T2, T3 plus notes in the last one). Never commit file by file.
8. **Tests.** You are not asked to write or run any test that queries the database. You only need: `npm run typecheck`, `npm run lint`, `npm run test:location`, `npm run build`. If you cannot run one, say so in your final report. Do not claim a command passed unless you ran it and saw it pass.
9. **Code comments.** Every changed block gets: `// CHANGED 2026-11-02 AI-STUDIO: <what and why>; see docs/HANDOFF.md`

2. Verified facts about the project

Fact	Where (VERIFIED 2 Nov 2026)
Stack: Next.js ^15.1, React ^19, TypeScript, Tailwind, @supabase/supabase-js, @turf/turf ^7.1, lucide-react; Node >= 20.	package.json
Vercel production build command is **bun run build**; it failed on main with exit 1 (lint_or_type_error).	Vercel deployment dpl_21ednuxTxpVtkiYPBu3n3jtPu1Vs

Fact	Where (VERIFIED 2 Nov 2026)
The live site is still on the older release (commit 96ff75d) and works; nothing here may break it.	Vercel alias iwacu-rwanda-2.vercel.app
The app is the folder **src/**. Old and imported code lives in **legacy/**, **national-grid/**, and our own map files in **public/rwanda-grid-map/**. On the branch fix/consolidation-build, tsconfig and ESLint already ignore legacy, national-grid and grid.	tsconfig.json (exclude), .eslintrc.json (ignorePatterns) on fix/consolidation-build
The map picker is **src/components/MapPinPicker.tsx** (296 lines). It loads Leaflet from https://unpkg.com/leaflet@1.9.4 and map tiles from https://{s}.tile.openstreetmap.org. Those are the two third-party runtime hosts to remove.	MapPinPicker.tsx lines 16-17 and 124
Our own map file exists on main and on the branch: **public/rwanda-grid-map/basemap/rwanda.pmtiles**, 45,127,285 bytes. Next to it: folders **demo** and **vendor**. The page URL of the tile file is **/rwanda-grid-map/basemap/rwanda.pmtiles**.	GitHub listing of public/rwanda-grid-map/basemap on main
A working demo page of the same map (MapLibre GL + PMTiles, no third-party host) exists. Its source is the model for T1.	folder public/rwanda-grid-map/basemap/demo/ ; the same page is the file basemap/demo/index.html (3,705 bytes) in the grid repo
Footer text now says: Inkomoko y'imipaka: National Institute of Statistics of Rwanda (CC BY 4.0). Ikarita: © OpenStreetMap contributors (ODbL).	src/app/layout.tsx line 41; test src/lib/independence.test.ts line 52
Calibration is loaded in line 1 of **src/lib/backoff.ts**: import calibrationConfig from "./calibration-v1.json". v1 is the identity curve, version "identity-v1", tau 0.8.	src/lib/backoff.ts line 1; src/lib/calibration-v1.json
The service worker **public/sw.js** (6,127 bytes) caches every successful same-origin GET in the shell cache (its last handler); CACHE_VERSION is "iwacu-v3-representative-points".	public/sw.js line 7 and lines 156-166

3. T1: our own map in MapPinPicker (the main task)

Goal. The map in `src/components/MapPinPicker.tsx` must be drawn from OUR file `/rwanda-grid-map/basemap/rwanda.pmtiles` with MapLibre GL and PMTiles, loaded from OUR domain. After the change, no request goes to unpkg.com or tile.openstreetmap.org. Everything the user sees and can do stays the same.

CHECK first. List the folder `public/rwanda-grid-map/basemap/vendor/`. Confirm these files exist: `maplibre-gl.js`, `maplibre-gl.css`, `pmtiles.js`, and the font folder `fonts/Noto Sans Regular/` with .pbf files. Also read `public/rwanda-grid-map/ATTRIBUTION.md` (828 bytes). If any of these is missing, STOP T1 and report which.

What the present component does (VERIFIED, keep all of it).

- Props: `onSelect(latitude, longitude, accuracyMeters)` and optional `initialPoint {latitude, longitude}`.
- Map opens when the user taps the button "Fungura ikarita"; state `mapStatus` is idle, loading, ready or error.
- Start view: the Rwanda overview centre latitude -1.94, longitude 29.87, Leaflet zoom 8. If `initialPoint` is given, or if localStorage key `iwacu-last-location-fix` holds a fix not older than 24 hours, start there at Leaflet zoom 14 and put the marker there.
- Limits: Leaflet minZoom 7, maxZoom 18; mouse-wheel zoom is OFF.
- Tapping the map sets the marker; the marker is draggable; the chosen point is shown as latitude, longitude with 4 decimals.
- Village borders: at Leaflet zoom 9 or more, 450 ms after the map stops moving, it calls `fetch("/api/location-map?" + params)` with west, south, east, north, cache force-cache, abortable, and draws the returned GeoJSON polygons: line colour #20603D, weight 1, opacity 0.8, fill #48A868, fill opacity 0.06. Below that zoom the borders are removed.
- Each polygon shows its name (`properties.name_kiny ?? properties.village`) as a tooltip, and tapping a polygon moves the marker to the polygon's bounding-box centre.
- The button "Koresha aha hantu" calls `onSelect(selected.latitude, selected.longitude, 200)` (the 200 is fixed, keep it exactly).
- All the Kinyarwanda messages (loading, error, border count, small print) stay as they are, except the attribution line below.

What to build.

1. Create `src/lib/basemap-style.ts` exporting `buildBasemapStyle(baseUrl: string)`. It holds the style in Appendix A (copy it exactly; its git blob SHA is `ba61cef7f714af3168de209ad1140bc55acca3ff` so the copy can be checked) and replaces every `__BASE__` with `baseUrl` (without a trailing slash). Call it with `window.location.origin + "/rwanda-grid-map/basemap"`.
2. In MapPinPicker, delete `loadLeaflet`, `LEAFLET_CSS`, `LEAFLET_JS`, the `window.L` declaration, and every Leaflet call. Add a small loader with the same pattern (one link tag, two script tags, no integrity attribute) that loads, from our own domain: `/rwanda-grid-map/basemap/vendor/maplibre-gl.css`, `/rwanda-grid-map/basemap/vendor/maplibre-gl.js`, `/rwanda-grid-map/basemap/vendor/pmtiles.js`. They define the globals `maplibregl` and `pmtiles` (the demo page uses exactly these).
3. Register the PMTiles protocol ONCE for the whole page (module-level flag, because React Strict Mode runs effects twice): `const protocol = new pmtiles.Protocol(); maplibregl.addProtocol("pmtiles", protocol.tile);`
4. Create the map: `new maplibregl.Map({ container, style: buildBasemapStyle(...), center: [lng, lat], zoom, minZoom, maxZoom })`, call `map.scrollZoom.disable()`, add `new maplibregl.NavigationControl()`, and add an attribution control with the text from the ATTRIBUTION.md you read in CHECK.
5. **Zoom numbers:** MapLibre zoom is one level lower than Leaflet zoom for the same view. Subtract 1 from every zoom number: 8 becomes 7, 14 becomes 13, minZoom 7 becomes 6, maxZoom 18 becomes 17, and the border threshold 9 becomes 8.
6. **Coordinates:** MapLibre takes [longitude, latitude]; the old code used [latitude, longitude]. Convert every place carefully.

7. Marker: `new maplibregl.Marker({ draggable: true }).setLngLat([lng, lat]).addTo(map)`; on `dragend` read `marker.getLngLat()` and update `selected`.
8. Borders: add a GeoJSON source with id `villages`, a fill layer `villages-fill` (fill-color #48A868, fill-opacity 0.06) and a line layer `villages-line` (line-color #20603D, line-width 1, line-opacity 0.8). Update with `map.getSource("villages").setData(geojson)`; below zoom 8 set empty data. Use `map.getBounds().getWest()/getSouth()/getEast()/getNorth()` for the four parameters.
9. **One click handler only.** In `map.on("click", ...)` first ask `map.queryRenderedFeatures(e.point, { layers: ["villages-fill"] })`. If a village polygon is under the tap: move the marker to the bounding-box centre of that polygon and show its name in a `maplibregl.Popup` (closeButton false). If not: move the marker to the tapped point. This keeps a polygon tap from being overridden by the map tap (a tooltip on hover does not exist on a phone, so the popup is shown on tap).
10. Cleanup: on close or unmount call `marker.remove()` and `map.remove()`, abort the pending fetch, clear the timer. After the map is shown call `map.resize()`.
11. Errors: if the three vendored files cannot load, set `mapStatus` to "error" (same message as now). If only tiles fail, keep the map, keep the borders, and let the user tap; show a short Kinyarwanda note.
12. Small print line at the bottom of the component: replace "Amakarita: © OpenStreetMap;" by "Amakarita: © OpenStreetMap, © OpenMapTiles;" and keep the rest of the sentence unchanged.
13. Footer in `src/app/layout.tsx`: keep the text and the exact substring `OpenStreetMap contributors (ODbL)`, and add the other credits that ATTRIBUTION.md requires, in the same sentence style.

Tests that exist today and what must happen to them (VERIFIED)

Test	What it asserts now	What you do
`src/lib/location-ui-contract.test.ts` lines 83-94	MapPinPicker source includes: `fetch("/api/location-map?"`, `layer.on("click"`, `onSelect(selected.latitude, selected.longitude, 200)`.	Keep the first and third strings as they are. The second is Leaflet-only: replace it with two assertions `mapPicker.includes("villages-fill")` and `mapPicker.includes("queryRenderedFeatures")`. Report this change in your final report.
`src/lib/independence.test.ts` line 52	Footer contains "OpenStreetMap contributors (ODbL)".	Keep it passing. Add a new test in the same file: MapPinPicker.tsx contains none of `unpkg.com`, `tile.openstreetmap.org`, `leaflet` (case-insensitive).
`npm run test:location`	Runs 10 test files with tsx and then node --test scripts/fit-calibration.test.mjs.	Must pass at the end. Never loosen an existing assertion to make it pass.

Done when (T1). (a) Searching `src/` for unpkg.com, tile.openstreetmap.org and leaflet finds nothing in MapPinPicker.tsx. (b) typecheck, lint, test:location and build pass, or you say honestly which you could not run. (c) Behaviour list above is unchanged.

4. T2: calibration v2 (measured on all of Rwanda)

Background (VERIFIED). On 2 November 2026 a national test sent 3,000 requests to the live service per seed using real test points from all 30 districts. Fitting on seed 424242 and measuring on the untouched seed 20261030 gave a calibration error (ECE, target 0.05 or lower) of: village 0.0173, cell 0.0123, sector 0.0154, district 0.0031, province 0.0014 (3,000 test requests per level). The fitted file is on the branch `wip/p4-results` at `results/calibration-v2.json` (version "isotonic-v2-national", tau 0.65580768).

1. Copy `results/calibration-v2.json` from branch `wip/p4-results` to `src/lib/calibration-v2.json` **byte for byte**. Do not retype or round any number. If you cannot copy the file exactly, SKIP T2 and say so.
2. In `src/lib/backoff.ts` change line 1 to import from `./calibration-v2.json`. Change nothing else in that file. Keep `src/lib/calibration-v1.json` in place: going back is a one-line change. If TypeScript complains about the cast on the imported file, cast through `unknown`.
3. Run `npm run test:location`. Tests that assert the old values ("identity-v1", tau 0.8, identity outputs) must be updated to the values **read from the new file**, never to numbers you guess. List every test you changed with the old and new expectation in your final report. Do NOT change any accuracy gate or threshold.

Done when (T2). The new file is identical to the source, backoff.ts imports it, and test:location passes (or you report exactly which test fails and why).

5. T3: service worker must leave the map file alone, T4: notes

Why (VERIFIED). The last handler of `public/sw.js` answers every same-origin GET with network-first and then stores any `response.ok` answer in the shell cache. A map file read in partial pieces (HTTP 206) cannot be stored, and the 45 MB file must never be copied into that cache.

1. In `public/sw.js`, right after the early return `if (request.method !== "GET" || url.origin !== self.location.origin) { return; }`, add: `if (url.pathname.startsWith("/rwanda-grid-map/") || request.headers.has("range")) { return; }` so the browser handles those requests itself.
2. Change CACHE_VERSION on line 7 from "iwacu-v3-representative-points" to "iwacu-v4-own-basemap" so old caches are replaced. CHECK first: search the tests and scripts for the old string and for sw.js; if something asserts it, update that assertion to the new string and report it.
3. **T4.** Append to `docs/HANDOFF.md` one entry in the format `[AI STUDIO] area: what - why - next` for each task. In `README.md` add a short section "One project" saying: src/ is the app; legacy/ is frozen old code; national-grid/ holds the grid work and is excluded from the build; public/rwanda-grid-map/ holds our own map files. Say nothing that is not true today.

6. NOT yours: parked on purpose (do not touch)

Item	Why it is parked (VERIFIED)
Overpass call in **src/app/api/location-context/route.ts** (line 95 calls https://overpass-api.de)	Replacing it needs a landmarks data set of our own. The extraction script exists in the grid repo but no landmark data is committed anywhere yet. Changing the route now would leave the Nearby places panel empty.
Offline engine (grid repo, packages/offline-locate and grid-lookup)	Needs the grid data files and the parity tests, which have not finished. Not wired into the app today.
Leaders side: src/lib/mock-data.ts (8 photo links on images.unsplash.com), src/components/AddLeaderModal.tsx line 53, mock fallback in src/app/api/leaders/route.ts	The mock data is used only when Supabase is not configured (route.ts line 49 and 182). It belongs to the separate leaders/governance build the owner starts next.
Database work, heavy validation jobs, security clean-up of the database	Done by other agents, with care for the free database.
AI assistant SDKs (@google/genai, groq-sdk in package.json)	The use of outside AI services is a separate decision.
Links to x.com, linkedin.com, wa.me	They are links the user may follow, not resources loaded by the page.

7. Your final report (required format)

Write one plain report, beginning with the words AI STUDIO. For each of T1, T2, T3 and T4 give: status (DONE, PARTIAL or SKIPPED), the files you changed, the exact commands you ran and what they printed, and anything you could not verify. Do not write "should work". If you skipped or changed anything that this order did not mention, say it. List the branch name and the commit hashes.

What happens next. Grok opens the pull request, watches the CI and the Vercel preview, and checks in a real browser that the page contacts only our own domain. Claude reads the result and decides about release. Nothing goes live without that decision.

Appendix A: the map style (exact copy, 2,821 bytes, git blob SHA ba61cef7f714af3168de209ad1140bc55acca3ff)

Save as the JSON template used by `src/lib/basemap-style.ts`. Keep the text `__BASE__` exactly; the function replaces it.

```
{
  "version": 8,
  "name": "iwacu-rwanda-light",
  "sources": {
    "rwanda": {
      "type": "vector",
      "url": "pmtiles://__BASE__/rwanda.pmtiles"
    }
  },
  "glyphs": "__BASE__/vendor/fonts/{fontstack}/{range}.pbf",
  "layers": [
    { "id": "bg", "type": "background", "paint": { "background-color": "#f0f4f0" } },
    { "id": "landcover", "type": "fill", "source": "rwanda", "source-layer": "landcover",
      "paint": { "fill-color": "#c6ddb0", "fill-opacity": 0.55 } },
    { "id": "landuse", "type": "fill", "source": "rwanda", "source-layer": "landuse",
      "paint": { "fill-color": "#d5e8c8", "fill-opacity": 0.4 } },
    { "id": "park", "type": "fill", "source": "rwanda", "source-layer": "park",
      "paint": { "fill-color": "#b8d9a0", "fill-opacity": 0.5 } },
    { "id": "water", "type": "fill", "source": "rwanda", "source-layer": "water",
      "paint": { "fill-color": "#7eb6d9" } },
    { "id": "waterway", "type": "line", "source": "rwanda", "source-layer": "waterway",
      "paint": { "line-color": "#7eb6d9", "line-width": 1 } },
    { "id": "building", "type": "fill", "source": "rwanda", "source-layer": "building", "minzoom": 13,
      "paint": { "fill-color": "#c9c4b8", "fill-opacity": 0.7 } },
    { "id": "transportation", "type": "line", "source": "rwanda", "source-layer": "transportation",
      "paint": { "line-color": "#f0c27a", "line-width": ["interpolate", ["linear"], ["zoom"], 8, 0.8, 14, 3] } },
    { "id": "transportation_name", "type": "symbol", "source": "rwanda", "source-layer": "transportation_name",
      "layout": { "text-field": ["get", "name"], "text-size": 11, "text-font": ["Noto Sans Regular"] },
      "paint": { "text-color": "#444", "text-halo-color": "#fff", "text-halo-width": 1 } },
    { "id": "boundary", "type": "line", "source": "rwanda", "source-layer": "boundary",
      "paint": { "line-color": "#888", "line-width": 1, "line-dasharray": [2, 2] } },
    { "id": "place", "type": "symbol", "source": "rwanda", "source-layer": "place",
      "layout": { "text-field": ["coalesce", ["get", "name"], ""], "text-size": 12, "text-font": ["Noto Sans Regular"] },
      "paint": { "text-color": "#222", "text-halo-color": "#fff", "text-halo-width": 1.2 } },
    { "id": "poi", "type": "symbol", "source": "rwanda", "source-layer": "poi", "minzoom": 12,
      "layout": { "text-field": ["get", "name"], "text-size": 10, "text-font": ["Noto Sans Regular"] },
      "paint": { "text-color": "#555", "text-halo-color": "#fff", "text-halo-width": 1 } },
    { "id": "mountain_peak", "type": "symbol", "source": "rwanda", "source-layer": "mountain_peak",
      "layout": { "text-field": ["get", "name"], "text-size": 11, "text-font": ["Noto Sans Regular"] },
      "paint": { "text-color": "#333", "text-halo-color": "#fff", "text-halo-width": 1 } }
  ]
}
```

Appendix B: working demo code for the loader order (model, VERIFIED)

```
<script src="../vendor/maplibre-gl.js"></script>
<script src="../vendor/pmtiles.js"></script>
...
const protocol = new pmtiles.Protocol();
maplibregl.addProtocol("pmtiles", protocol.tile);
const base = new URL("..", window.location.href).href.replace(/\/$/, "");
const tilesUrl = base + "/rwanda.pmtiles";
const glyphs = base + "/vendor/fonts/{fontstack}/{range}.pbf";
... new maplibregl.Map({ container: "map", style, center: [30.06, -1.95], zoom: 11 });
map.addControl(new maplibregl.NavigationControl(), "bottom-right");
```

Source: the grid repo file basemap/demo/index.html (3,705 bytes), shipped to this repository in the folder public/rwanda-grid-map/basemap/demo/ (CHECK the file name there).